

Chapter 7

Event Selection

The event selection separates the four signal topologies introduced in Section 3.1.5 according to the presence of a b -tagged jet and the kinematics of resonant or non-resonant production. The analysis variables and common preselection are introduced first, followed by the optimisation of the one- b -jet regions the definition of the zero- b -jet regions, and the contribution of $\tau\tau$ signals to the selected final state.

7.1 Preselection and variables definition

The analysis variables are constructed from the calibrated objects defined in Chapter 6, after overlap removal. Throughout this chapter, τ denotes the reconstructed visible decay products of the selected hadronically decaying τ -lepton. For signal events, the neutrino from the τ decay is not included in this four-momentum and contributes to $\vec{p}_T^{\text{miss}}$ together with the neutrino produced in the hard scattering.

7.1.1 Analysis variables

The transverse momenta, angular separations and reconstructed masses characterise the energy scale and geometry of the $\tau\nu + \text{jets}$ final state. They distinguish the signal from $W + \text{jets}$ and top-quark production, and separate the resonant and non-resonant signal contributions.

The τ transverse momentum and E_T^{miss} measure the visible and invisible transverse activity in the event. Heavy-leptoquark production can produce an energetic τ -lepton and neutrinos, whereas most SM events populate lower values of these quantities. Requiring large p_T^τ and E_T^{miss} therefore selects the high-energy part of the $\tau\nu$ spectrum in which a leptoquark contribution can become distinguishable [1]. The two quantities need not be equal because the jets carry transverse momentum and the τ -decay neutrino contributes to $\vec{p}_T^{\text{miss}}$.

The azimuthal separation $\Delta\phi(\tau, E_T^{\text{miss}})$ is the angle between the visible τ momentum and $\vec{p}_T^{\text{miss}}$ in the transverse plane, with values between zero and π . In non-resonant production with a relatively soft recoiling jet, the τ and the invisible system tend to be back-to-back. A lower requirement on $\Delta\phi(\tau, E_T^{\text{miss}})$ favours this geometry and reduces top-quark backgrounds with a more complex final state. Resonant production contains an energetic jet from the leptoquark decay, so a back-to-back requirement on the τ and $\vec{p}_T^{\text{miss}}$ can also remove the signal of interest.

The transverse mass m_T combines the two transverse momentum scales with their angular separation:

$$m_T = \sqrt{2p_T^\tau E_T^{\text{miss}} [1 - \cos\Delta\phi(\tau, E_T^{\text{miss}})]}. \quad (7.1)$$

This definition neglects the small visible τ mass. It is useful because the longitudinal momenta of the neutrinos are unmeasured and the full $\tau\nu$ invariant mass cannot be reconstructed. For an on-shell $W \rightarrow \tau\nu$ decay with correctly reconstructed momenta and no other source of $\vec{p}_T^{\text{miss}}$, m_T is bounded by the W -boson mass. Off-shell W production, additional neutrinos and detector resolution populate the high- m_T tail of the SM background. A heavy virtual leptoquark can enhance this tail, making m_T particularly useful for non-resonant production. Equation 7.1 also shows that its selection is correlated with those on p_T^τ , E_T^{miss} and $\Delta\phi(\tau, E_T^{\text{miss}})$.

The visible invariant masses $m_{b\tau}$ and $m_{j\tau}$ are reconstructed from the selected b -jet or leading jet and the visible τ :

$$m_{b\tau}^2 = (p_b + p_\tau)^2, \quad m_{j\tau}^2 = (p_{j_1} + p_\tau)^2, \quad (7.2)$$

where p_b , p_{j_1} and p_τ are four-momenta, and j_1 is the jet with the largest p_T . In resonant $U_1 \rightarrow b\tau$ production, $m_{b\tau}$ retains information about the leptoquark mass. The neutrino from the τ decay shifts the visible mass below M_U and broadens its distribution, with further smearing from the reconstructed jet and τ energies. A lower mass threshold therefore selects heavy-particle decays without requiring a narrow peak at M_U . The analogous variable $m_{j\tau}$ is used in the zero- b -jet regions, where the leading jet can originate from $U_1 \rightarrow s\tau$.

The selected-jet transverse momentum distinguishes the two production topologies. The b -jet from a resonant $U_1 \rightarrow b\tau$ decay is typically energetic because it shares the decay energy of a heavy particle. A b -jet radiated in non-resonant production is typically softer. The b -jet p_T is therefore used to divide the one- b -jet sample, while the leading-jet p_T is used in the zero- b -jet sample. These reconstructed categories enrich the corresponding production modes without separating them exactly.

The object multiplicities specify the reconstructed final state. The numbers of hadronic τ candidates, electrons, muons, jets and b -tagged jets are denoted by N_τ , N_e , N_μ , N_j and N_b , respectively, with $N_\ell = N_e + N_\mu$. All jets, including b -tagged jets, enter N_j , so $N_j - N_b$ counts

jets that are not b -tagged. An upper requirement on this multiplicity suppresses $t\bar{t}$ events, whose two top-quark decays often produce several jets.

7.1.2 Multijet suppression

Multijet events can enter the selected sample when a jet is misidentified as a hadronic τ and a mismeasured jet produces large E_T^{miss} . In particular, an underestimated jet momentum produces a momentum imbalance pointing towards that jet. The minimum azimuthal separation and the corresponding momentum ratio are therefore defined as

$$\Delta\phi_{\min} = \min_j \Delta\phi(\vec{p}_T^{\text{miss}}, \vec{p}_T^j), \quad R_{\text{miss}/j} = \frac{E_T^{\text{miss}}}{p_T^{j_{\min}}}, \quad (7.3)$$

where $j_{\min}$ is the jet with the smallest azimuthal separation from $\vec{p}_T^{\text{miss}}$. It need not be the leading jet. Mismeasured multijet events tend to have small $\Delta\phi_{\min}$ and a modest $R_{\text{miss}/j}$. By contrast, the neutrinos in signal events can produce large E_T^{miss} even when a much softer jet happens to lie nearby in azimuth.

Events are retained if

$$\Delta\phi_{\min} > 0.4 \quad \text{or} \quad R_{\text{miss}/j} > 6. \quad (7.4)$$

The second condition retains signal events that would be lost with the angular requirement alone. In the simulated study described in Appendix B, this selection rejects approximately 86% of the multijet events while retaining 96.4% of the signal after the loose study selection. These efficiencies describe the cleaning requirement at that stage, before the full SR selections.

7.1.3 Common preselection

The preselection establishes the $\tau\nu + \text{jets}$ final state before the kinematic optimisation. Events must pass the lowest-threshold unprescaled E_T^{miss} trigger for the relevant data-taking period, contain exactly one selected hadronic τ candidate, and contain no selected electron or muon. The multijet suppression in Eq. 7.4 is applied at this stage. The object definitions and data-quality requirements are those already given in Chapters 6 and 5. The baseline for the one- b -jet optimisation additionally requires $N_b = 1$ with $p_T^b > 50$ GeV and $E_T^{\text{miss}} > 200$ GeV. The latter requirement selects the trigger-efficiency plateau discussed in Section 5.4. The zero- b -jet category replaces the tagged jet with at least one untagged jet, as described in Section 7.3. The baseline requirements and final kinematic selections are collected together in Table 7.3.

7.2 One b-jet signal region (SR1b) optimisation and definition

The SR1b regions select exactly one b -tagged jet to target the $b\tau$ interaction in the U_1 model. The non-resonant selection is motivated by the high- m_T $\tau\nu + b$ strategy of Ref. [1]: an energetic τ and large E_T^{miss} select the hard scattering, the b -tag reduces the inclusive W +jets background, and a restriction on additional jets suppresses top-quark production. The analysis also retains a resonant category to select an on-shell leptoquark decaying into $b\tau$. The sample is divided at $p_T^b = 250$ GeV, with $p_T^b > 250$ GeV defining SR1b-Res and $50 < p_T^b < 250$ GeV defining SR1b-NonRes.

7.2.1 Scan variables and optimisation criterion

The kinematic thresholds are scanned together because the discriminating variables are correlated. For example, increasing E_T^{miss} also raises m_T at fixed p_T^τ and $\Delta\phi(\tau, E_T^{\text{miss}})$, so the effect of an additional m_T requirement depends on the other cuts. Lower thresholds on p_T^τ , E_T^{miss} and m_T are scanned in both categories, together with an upper limit on $N_j - N_b$. A lower threshold on $\Delta\phi(\tau, E_T^{\text{miss}})$ is tested for the non-resonant topology. For the resonant topology, upper thresholds on this angle and a lower threshold on $m_{b\tau}$ are tested. The complete scan grid is given in Table 7.1.

Table 7.1 Thresholds tested in the SR1b optimisation. Each set lists the trial values for the indicated inequality. A zero mass threshold or an angular upper threshold of π represents no effective requirement on that variable.

Variable	SR1b-Res	SR1b-NonRes
$\Delta\phi(\tau, E_T^{\text{miss}})$	$< \{0.3, 1.2, 2.4, 2.6, \pi\}$	$> \{0.3, 1.2, 2.4, 2.6\}$
$m_{b\tau}$ [GeV]	$> \{0, 800\}$	Not scanned
E_T^{miss} [GeV]	$> \{200, 250, 300, 350, 400, 500, 600\}$	
p_T^τ [GeV]	$> \{200, 250, 300, 350, 400, 500, 600\}$	
m_T [GeV]	$> \{100, 200, 300, 400, 500, 600, 700, 800, 900, 1000\}$	
$N_j - N_b$		$\leq \{1, 2, 3\}$

The mass scan uses only $m_{b\tau} > 0$ and 800 GeV. The visible mass distribution is broad because of the unmeasured τ -decay neutrino, while the simulated background has limited statistics in its high-mass tail. These features favour an inclusive mass requirement over a finely scanned mass window. The resonant optimisation targets $M_U = 1.5$ TeV, where on-shell production is important. The non-resonant optimisation uses $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$, where virtual exchange provides the dominant signal contribution.

The expected sensitivity of each cut combination is ranked using an approximate counting-experiment significance [2]:

$$Z = \sqrt{2 \left[n \ln \left(\frac{n(b + \sigma_b^2)}{b^2 + n\sigma_b^2} \right) - \frac{b^2}{\sigma_b^2} \ln \left(\frac{b^2 + n\sigma_b^2}{b(b + \sigma_b^2)} \right) \right]}, \quad (7.5)$$

where s and b are the expected signal and background event yields, $n = s + b$, and σ_b is the absolute uncertainty in the background expectation. A fractional background uncertainty of 30%, corresponding to $\sigma_b = 0.30b$, is assumed for this scan. The scan uses the pure BSM signal yield. The effect of interference on the final signal prediction is discussed below. The quantity Z is used to compare selections, while the final confidence limits are obtained with the likelihood analysis in Chapter 10.

Cut combinations with $b \leq 2$ are discarded to avoid selecting apparently background-free regions caused by limited MC statistics. If a particular simulated background has a negative weighted yield after a trial selection, its contribution is set to zero when evaluating the scan criterion. This is a prescription for comparing trial cuts in sparse simulated samples. It does not imply a negative physical event rate.

7.2.2 Interference and signal efficiency

The interference term in Eq. 5.4 changes the signal contribution remaining after the selection. If S denotes the pure BSM yield and I the interference yield, the excess above the SM prediction is $S + I$ at the nominal signal normalisation. Destructive interference gives $I < 0$, so a selection ranked using S alone can overestimate the final sensitivity. Its effect is therefore checked after the optimisation and included in the final statistical model.

The requirements on p_T^τ , E_T^{miss} and m_T suppress the low-energy phase space in which interference with SM W production is most important. The b -jet requirement also reduces the relevant SM amplitudes, for which charged-current transitions involving a b quark are suppressed by the small CKM elements V_{cb} and V_{ub} . In particular, the non-resonant requirements $p_T^\tau > 200$ GeV, $E_T^{\text{miss}} > 400$ GeV and $m_T > 600$ GeV retain the hard tail while reducing the destructive contribution. The residual interference has a negligible effect on the sensitivity of the one- b -jet regions in the parameter range probed by the analysis. This conclusion does not apply to the zero- b -jet regions, where the interference contribution must be retained, as discussed in Section 7.3.

The signal efficiency is monitored through the cumulative cutflow in Table 7.2. For $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$, the resonant selection retains 4.2 of the 35.2 expected signal events passing the one- b -jet baseline, or approximately 12%. For the non-resonant

optimisation benchmark, it retains 6.6 of 111.0 events, or approximately 6%. The background decreases from approximately 4027 baseline events to 2.4 and 2.1 events, respectively. These fractions describe signal retention during the cut optimisation and use the baseline sample as their denominator. The acceptance times selection efficiency, $A\epsilon$, instead uses the total generated signal as its reference and includes the kinematic acceptance and reconstruction and selection losses. The cutflow uses the pure BSM contribution because the signed interference yield is not an event-selection efficiency. The tabulated numbers describe the optimisation study before the background fit, while the final selections are chosen with the statistical uncertainty in the background prediction kept below 30%.

Table 7.2 Cumulative signal (S) and background (B) yields in the one- b -jet optimisation. The resonant and non-resonant signal benchmarks are $(M_U [\text{TeV}], g_U, \beta_L^{23}) = (1.5, 1.5, 0.6)$ and $(2.5, 2.5, 1.0)$, respectively. These are simulated expectations before the background fit.

Resonant			Non-resonant		
Cumulative requirement	S	B	Cumulative requirement	S	B
Baseline	35.2	4027.3	Baseline	111.0	4027.3
$p_T^\tau > 200 \text{ GeV}$	19.9	193.1	$p_T^\tau > 200 \text{ GeV}$	70.5	193.1
$E_T^{\text{miss}} > 200 \text{ GeV}$	19.9	193.1	$E_T^{\text{miss}} > 400 \text{ GeV}$	26.4	15.1
$m_T > 200 \text{ GeV}$	19.4	143.6	$m_T > 600 \text{ GeV}$	24.0	6.4
$m_{b\tau} > 800 \text{ GeV}$	4.9	5.2	$\Delta\phi(\tau, E_T^{\text{miss}}) > 1.2$	23.8	6.4
$N_j - N_b \leq 3$	4.7	4.3	$N_j - N_b \leq 1$	17.2	2.8
$p_T^b > 250 \text{ GeV}$	4.2	2.4	$50 < p_T^b < 250 \text{ GeV}$	6.6	2.1

7.2.3 Selected regions and background composition

The selected SR1b-Res requirements are $p_T^\tau > 200 \text{ GeV}$, $E_T^{\text{miss}} > 200 \text{ GeV}$, $m_T > 200 \text{ GeV}$ and $m_{b\tau} > 800 \text{ GeV}$, with at most three jets that are not b -tagged. No additional $\Delta\phi(\tau, E_T^{\text{miss}})$ requirement is retained. The visible-mass requirement provides strong rejection of backgrounds without a heavy $b\tau$ resonance, while the energetic b -jet selects the two-body decay topology. The region is treated as a single event-counting bin because the available background simulation does not support a finely binned fit to the broad $m_{b\tau}$ distribution.

The selected SR1b-NonRes requirements are $p_T^\tau > 200 \text{ GeV}$, $E_T^{\text{miss}} > 400 \text{ GeV}$, $m_T > 600 \text{ GeV}$ and $\Delta\phi(\tau, E_T^{\text{miss}}) > 1.2$, with at most one jet that is not b -tagged. The tighter E_T^{miss} and m_T requirements select the high-energy tail, and the smaller allowed jet multiplicity reduces the top-quark background. No $m_{b\tau}$ requirement is imposed because the b -jet and τ do not reconstruct an on-shell leptoquark in the target topology. This region also contributes a single event count to the fit.

The loose $N - 1$ distributions in Figures 7.1 and 7.2 illustrate the variables entering the optimisation. An $N - 1$ distribution omits the requirement on the plotted variable while retaining the other requirements. To make the shapes visible with the available simulation, the resonant m_T and p_T^τ thresholds are lowered by 100 GeV. For the non-resonant region, the m_T , E_T^{miss} and p_T^τ thresholds are each lowered by 100 GeV before the plotted requirement is removed. The trigger-plateau requirement $E_T^{\text{miss}} > 200$ GeV is retained. These distributions therefore illustrate a larger phase space than the final SRs. The visible-mass distribution extends below the true leptoquark mass because the τ -decay neutrino is unmeasured, while the non-resonant signal extends into the high- m_T and high- E_T^{miss} tails.

The remaining background in both one- b -jet regions is dominated by W +jets, principally $W \rightarrow \tau\nu$ with a hadronic τ decay. The tagged jet can originate from heavy-flavour production or a misidentified light-flavour jet. In the final background prediction, after the control-region constraints described in Chapter 8, W +jets contributes approximately 70% in each region. In SR1b-Res, diboson production contributes about 15%, followed by $t\bar{t}$ and single-top production, including Wt . In SR1b-NonRes, $t\bar{t}$ contributes about 14%, $Z \rightarrow \tau\tau$ +jets about 9%, and diboson production about 5%. Top-quark events can pass the selection through $t\bar{t} \rightarrow b\tau\nu\bar{b}\tau\nu$ or $Wt \rightarrow b\tau\nu\tau\nu$ when only one τ is selected and the remaining decay products satisfy the jet and lepton requirements. The $Z \rightarrow \tau\tau$ contribution similarly requires only one selected τ , with neutrinos from the decays producing E_T^{miss} . Among events with a jet misidentified as a τ , $Z \rightarrow \nu\nu$ +jets is the main source because the invisible Z decay supplies genuine missing momentum. Multijet production is negligible after the cleaning and full SR requirements.

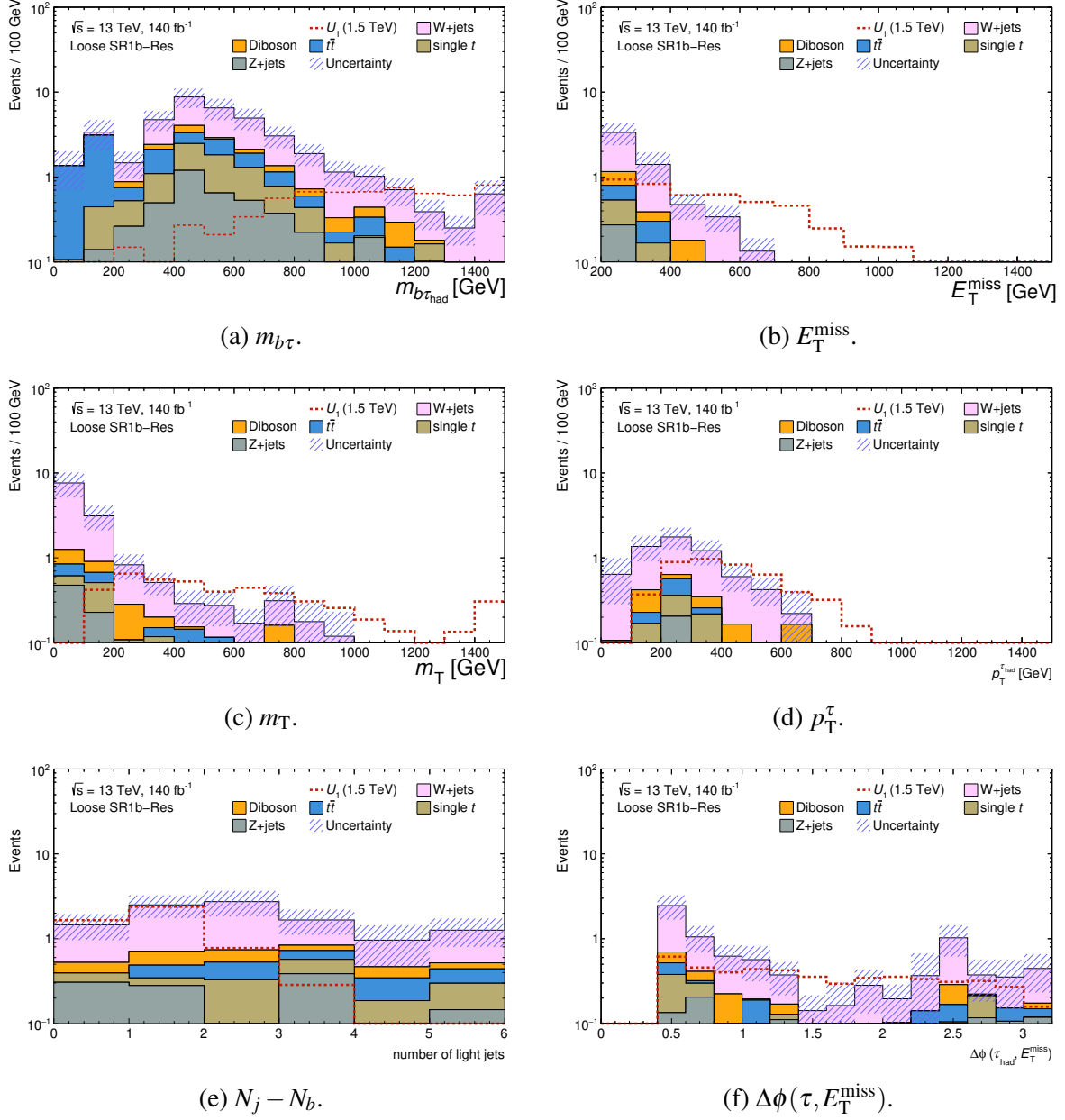


Fig. 7.1 Distributions of the six scan variables in the loose resonant one- b -jet region. The m_T and p_T^τ thresholds are reduced to 100 GeV, and the requirement on each plotted variable is removed, retaining the baseline $E_T^{\text{miss}} > 200$ GeV. The stacked backgrounds are shown before the fit. The dotted signal curve is the pure BSM prediction at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. The hatched band represents the background uncertainty. The plot notation τ_{had} denotes the visible τ candidate.

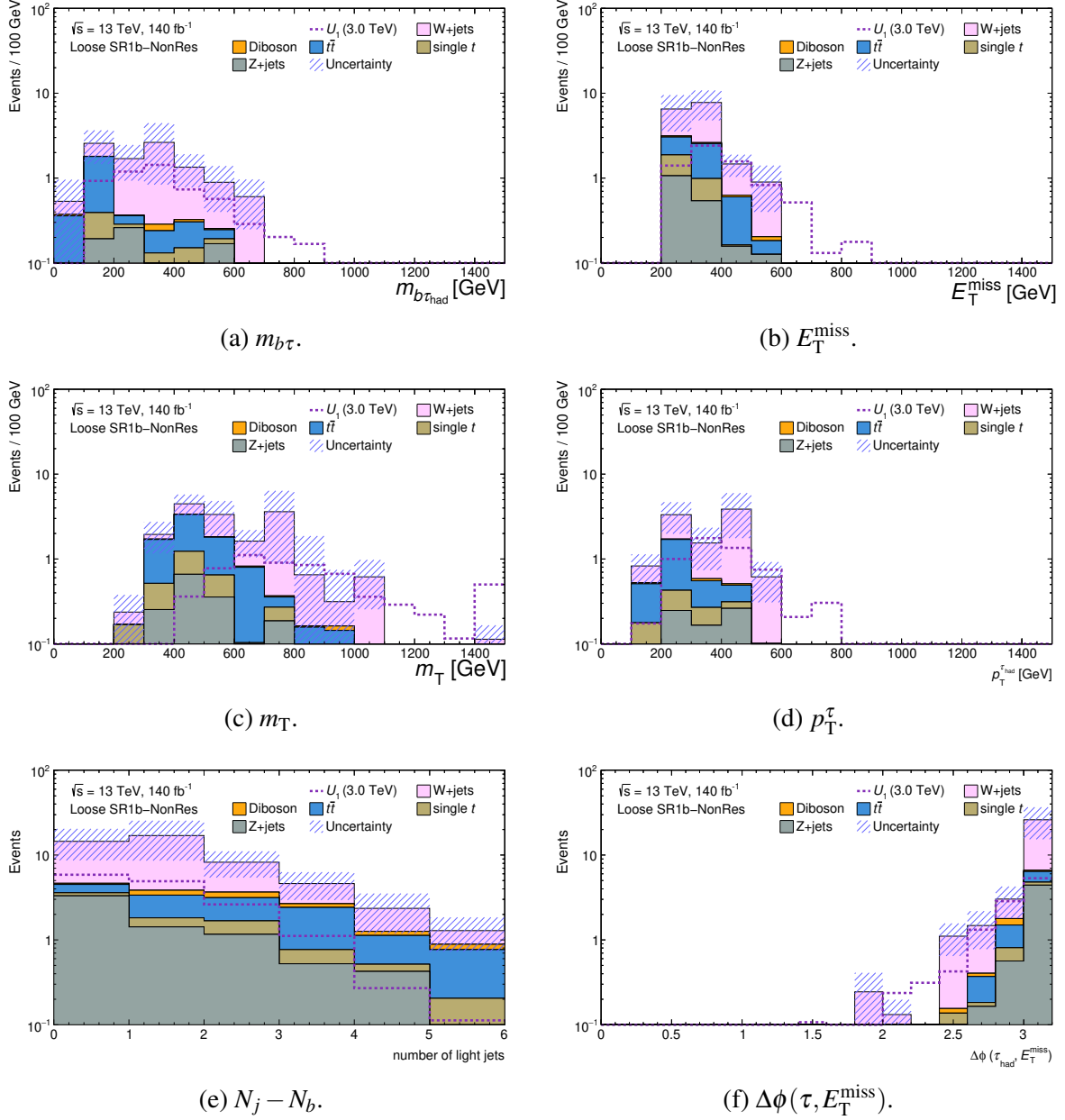


Fig. 7.2 Distributions of the six scan variables in the loose non-resonant one- b -jet region. Before removing the plotted requirement, the thresholds are lowered to $m_T > 500$ GeV, $E_T^{\text{miss}} > 300$ GeV and $p_T^\tau > 100$ GeV. The baseline $E_T^{\text{miss}} > 200$ GeV is retained. The stacked backgrounds are shown before the fit, and the hatched band represents their uncertainty. The dotted signal curve is the pure BSM prediction at $(M_U, g_U, \beta_L^{23}) = (3.0 \text{ TeV}, 2.5, 1.0)$.

7.3 b -veto signal region definition (SR0b)

The SR0b regions select signal events with no reconstructed b -tagged jet and extend the sensitivity to large β_L^{23} . As shown by the flavour current and coupling texture in Eqs. 3.6 and 3.7, β_L^{23} controls the $s\tau$ interaction and contributes $V_{cs}\beta_L^{23}$ to the $c\nu_\tau$ interaction. The resulting zero- b -jet topologies were introduced in Section 3.1.5. In particular, $U_1 \rightarrow s\tau$ produces a resonant signal with a genuine non- b jet, so SR0b retains a physically distinct signal contribution in addition to events in which a b -jet is not tagged or reconstructed. A leading jet in this category is not explicitly identified as a strange-quark jet. The truth-level study also identifies a substantial contribution with a leading c jet. At $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$, approximately 30% and 60% of the selected truth-level resonant-category events have a leading s or c jet, respectively. The s -jet contribution contains the $s\tau$ resonance, whereas the c -jet contribution can contain an on-shell $U_1 \rightarrow c\nu_\tau$ decay. For the latter, the resonance is in the $c\nu_\tau$ system rather than the visible jet- τ mass. In the non-resonant truth-level category, approximately 33% and 11% of the events have a radiated leading s or c jet, with most of the remainder containing a gluon jet in $sc \rightarrow \tau\nu_\tau$ production. These fractions describe the chosen benchmark and truth selection, rather than a flavour measurement of the reconstructed jets.

The two SR0b regions are defined by replacing the one- b -jet requirement with $N_b = 0$ and using the leading jet in the topology-dependent selections. The resonant region requires $p_T^{j_1} > 250 \text{ GeV}$ and $m_{j\tau} > 800 \text{ GeV}$, while the non-resonant region requires $50 < p_T^{j_1} < 250 \text{ GeV}$. The p_T^τ , E_T^{miss} , m_T and $\Delta\phi(\tau, E_T^{\text{miss}})$ thresholds follow the corresponding one- b -jet regions. The upper limits are imposed on the total jet multiplicity: $N_j \leq 4$ for SR0b-Res and $N_j \leq 2$ for SR0b-NonRes. These preserve the total jet limits used in SR1b after replacing the tagged jet with an untagged leading jet.

The $E_T^{\text{miss}} > 200 \text{ GeV}$ requirement in SR0b-Res is retained to keep its kinematic selection aligned with SR1b-Res. A supplementary study of tighter E_T^{miss} thresholds is presented in Appendix C. The SR0b-NonRes threshold remains $E_T^{\text{miss}} > 400 \text{ GeV}$, which also maximises the significance in that study.

Signal efficiency in SR0b is evaluated for the pure BSM contribution, with $A\epsilon$ defined as in Section 7.2.2. For illustration, at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.0, 0.2)$ with $\beta_R^{33} = 0$, $A\epsilon$ is 0.30% and 0.13% in SR0b-Res and SR0b-NonRes, compared with 0.64% and 0.069% in the corresponding SR1b regions. These efficiencies depend on the mass and flavour couplings because the relative contributions of the production and decay channels change. The destructive interference in the final zero- b -jet selections reduces the nominal pure BSM yield by approximately 20% at $M_U = 1.5 \text{ TeV}$ and by about 30–40% at higher masses. This reduction is appreciable even after the high- m_T and E_T^{miss} requirements, so the final prediction includes

the signed interference contribution explicitly. The supplementary E_T^{miss} scan in Appendix C also shows that a higher E_T^{miss} threshold can suppress this contribution.

Figure 7.3 shows the visible-mass distribution before the mass requirement in SR0b-Res. It also shows the transverse-mass distribution above 600 GeV in SR0b-NonRes. The high- $m_{j\tau}$ requirement selects the visible products of a heavy resonance, while the high- m_T requirement selects non-resonant production. Together with the one- b -jet regions, these selections retain sensitivity to both $b\tau$ and $s\tau$ final states over the coupling range studied in this thesis.

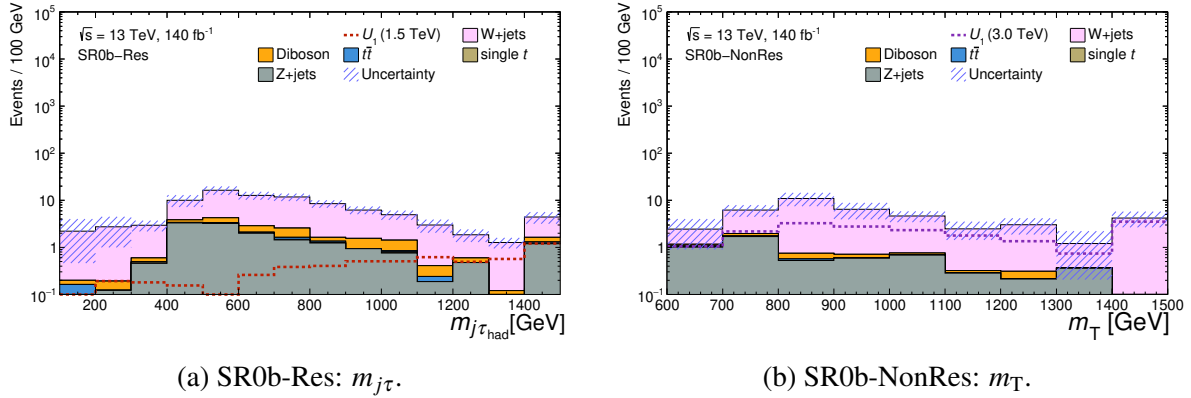


Fig. 7.3 Distributions of (a) $m_{j\tau}$ in SR0b-Res with the mass requirement removed and (b) m_T in SR0b-NonRes above 600 GeV. The other SR requirements are retained. The backgrounds are shown before the fit, and the hatched band represents their uncertainty. The dotted curves show pure BSM signals at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$ in (a) and $(3.0 \text{ TeV}, 2.5, 1.0)$ in (b).

The zero- b -jet background is dominated by $W \rightarrow \tau\nu$ +jets, since vetoing tagged b jets removes much of the top-quark background while retaining W production with light-flavour jets. In the final prediction after the control-region constraints, W +jets contributes approximately 76% in SR0b-Res and 82% in SR0b-NonRes. In the resonant region, diboson production contributes about 10%, $Z \rightarrow \nu\nu$ +jets about 8%, and $Z \rightarrow \tau\tau$ +jets about 5%. In the non-resonant region, the corresponding contributions are approximately 3%, 6% and 8%. The $Z \rightarrow \nu\nu$ events contain a jet misidentified as a τ , while $Z \rightarrow \tau\tau$ and diboson events can contain genuine hadronic τ decays and neutrinos. The combined $t\bar{t}$ and single-top contribution is below 1% in each zero- b -jet region.

The final requirements for all four mutually exclusive SRs are summarised in Table 7.3. Each region contributes one event count to the simultaneous statistical analysis.

Table 7.3 Common baseline requirements and final signal-region selections. The object definitions are given in Chapter 6. A dash denotes no additional requirement. The jet multiplicity includes b -tagged jets.

Baseline requirements				
Trigger	Lowest-threshold unscaled E_T^{miss} trigger			
N_τ	1			
$N_\ell = N_e + N_\mu$	0			
Multijet suppression	$\Delta\phi_{\min} > 0.4$ or $R_{\text{miss}/j} > 6$			
Requirement	Resonant		Non-resonant	
	SR1b-Res	SR0b-Res	SR1b-NonRes	SR0b-NonRes
N_b	1	0	1	0
p_T^b [GeV]	> 250	–	$50 < p_T^b < 250$	–
$p_T^{j_1}$ [GeV]	–	> 250	–	$50 < p_T^{j_1} < 250$
p_T^τ [GeV]			> 200	
E_T^{miss} [GeV]	> 200			> 400
m_T [GeV]	> 200			> 600
$m_{b\tau}$ [GeV]	> 800	–	–	–
$m_{j\tau}$ [GeV]	–	> 800	–	–
$\Delta\phi(\tau, E_T^{\text{miss}})$		–		> 1.2
N_j		≤ 4		≤ 2

7.4 Tau-pair signal contamination

Leptoquark production with a $\tau\tau$ final state can also populate the $\tau\nu$ + jets signal regions. Although two τ -leptons are produced, an event can satisfy the single- τ requirement when only one is reconstructed and selected. Neutrinos from the τ decays produce missing momentum. This is an additional BSM contribution from the same U_1 model, rather than an SM background.

The study uses simulated $\tau\tau$ samples generated with $\beta_L^{33} = 1$, $\beta_R^{33} = 0$ and $\beta_L^{23} = 0$. Their selected yields are compared with available $\tau\nu$ samples near the exclusion sensitivity, at $(M_U, g_U, \beta_L^{23}) = (1.8 \text{ TeV}, 3.0, 0.6)$ and $(3.0 \text{ TeV}, 2.0, 1.4)$. The corresponding $\tau\tau$ samples use $g_U = 2.8$ and 2.1 . The $\tau\tau$ production cross section is recalculated with MADGRAPH5_AMC@NLO at larger β_L^{23} , and the selected yields are scaled by the cross-section ratio to their generation point. This estimates the additional rate while retaining the acceptance and kinematic shapes of the available samples. It therefore provides a sensitivity study rather than a simulation of the full coupling dependence of the $\tau\tau$ acceptance.

The estimated additional yield is of order 35% of the nominal $\tau\nu$ yield in the one- b -jet regions and several percent in the zero- b -jet regions. To assess the effect on the search, the nominal signal predictions are increased by the factors shown in Table 7.4, and the exclusion fit is repeated.

Table 7.4 Additional signal normalisation used to estimate the sensitivity to $\tau\tau$ production. The factors multiply the nominal $\tau\nu$ signal prediction in the supplementary fit.

Region	Additional contribution	Applied factor
SR1b-Res	37%	1.37
SR1b-NonRes	33%	1.33
SR0b-Res	5%	1.05
SR0b-NonRes	7%	1.07

The expected upper limit on the signal strength improves by approximately 8–14% in this test. The corresponding change in the coupling limit is smaller because the signal rate depends strongly on g_U . At $M_U = 1.8 \text{ TeV}$ and $\beta_L^{23} = 0.6$, the expected upper limit on g_U changes from 2.18 to 2.07, an improvement of approximately 5%. At $M_U = 3.0 \text{ TeV}$ and $\beta_L^{23} = 1.4$, it changes from 1.96 to 1.91, or approximately 2.6%. The nominal interpretation retains the $\tau\nu$ signal model. The supplementary study indicates that the omitted positive $\tau\tau$ contribution could strengthen the expected coupling constraints by a few percent under the assumptions used for the rate estimate.

Appendix B

Multijet background study

The multijet suppression used in Section 7.1.2 is studied with simulated dijet and leptoquark events. The purpose is to identify a selection that removes events with instrumental E_T^{miss} while retaining the signal with neutrinos. The multijet production cross section is large, so a small probability for a jet to be misidentified as a τ , combined with a large jet-energy mismeasurement, can produce events with the selected final state.

B.1 Origin of the discriminating variables

Jet-momentum mismeasurement produces a characteristic direction for the apparent missing momentum. For example, if the transverse momentum of one jet is underestimated by $\delta \vec{p}_T^j$, the reconstructed momentum imbalance receives a contribution in the direction of the missing part of that jet momentum. The resulting $\vec{p}_T^{\text{miss}}$ is therefore approximately aligned with the jet. In a signal event, the neutrinos produce real missing momentum, whose direction is not fixed by a jet-energy mismeasurement.

The variables $\Delta\phi_{\text{min}}$ and $R_{\text{miss}/j}$ defined in Eq. 7.3 test these two features. The first measures the angular alignment, and the second compares the missing momentum with the momentum of the closest jet in azimuth. The combination is useful because a low- p_T jet can lie close to the direction of real missing momentum in a signal event. Such an event can have small $\Delta\phi_{\text{min}}$ but large $R_{\text{miss}/j}$.

B.2 Study selection and distributions

The study selection is kept loose to retain enough simulated multijet events to examine the relevant distributions. It requires the E_T^{miss} trigger, exactly one hadronic τ candidate, no selected

electron or muon, and at least one jet, as listed in Table B.1. The final SR thresholds on E_T^{miss} , p_T^τ , m_T and the visible invariant mass are not imposed at this stage. No b -tag multiplicity is required.

Table B.1 Loose selection used to study the multijet suppression. This selection defines the denominator of the rejection fractions shown below.

Requirement	Selection
Trigger	Lowest-threshold unprescaled E_T^{miss} trigger
Hadronic τ multiplicity	$N_\tau = 1$
Electron and muon veto	$N_\ell = 0$
Jet multiplicity	$N_j \geq 1$

Figure B.1 shows the two-dimensional distributions after this selection. The multijet sample is concentrated at small $\Delta\phi_{\text{min}}$ and relatively small $R_{\text{miss}/j}$. The signal occupies a broader angular range and extends to larger momentum ratios. Approximately 90% of the multijet events have $R_{\text{miss}/j} < 6$, compared with approximately 55% of the signal in the study sample. The ratio requirement alone would therefore reject a substantial fraction of the signal. It is instead combined with the angular requirement so that either a large angle or a large ratio is sufficient to retain an event.

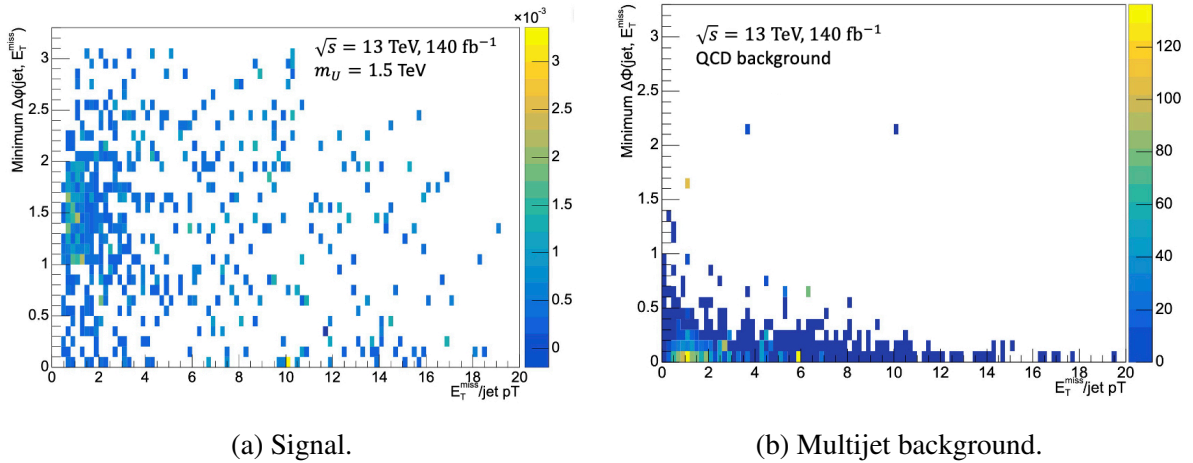


Fig. B.1 Correlation between the minimum jet- $\vec{p}_T^{\text{miss}}$ azimuthal separation and the ratio of E_T^{miss} to the p_T of the closest jet in azimuth, for (a) simulated signal with $M_U = 1.5 \text{ TeV}$ and (b) simulated multijet background after the loose study selection. The concentration of multijet events at small values of both variables motivates a combined requirement.

B.3 Threshold scan and signal retention

The angular threshold is scanned while the momentum-ratio threshold is fixed at six. For an angular threshold X , the rejected fraction is defined as

$$\text{Rejection ratio} = \frac{N(\Delta\phi_{\min} < X \text{ and } R_{\text{miss}/j} < 6)}{N(\text{all events after preselection})}. \quad (\text{B.1})$$

The numerator requires both small angular separation and a small momentum ratio. Its complement is the accepted region, defined by the logical OR in Eq. 7.4. The rejected fraction is evaluated separately for signal and multijet events, using each sample's yield after the loose selection as its denominator.

Figure B.2 shows the resulting rejection fractions. At $X = 0.4$, approximately 86% of the multijet events are rejected, while the signal loss is 3.6%. The corresponding signal efficiency for the cleaning cut is therefore $1 - 0.036 = 96.4\%$. Increasing the angular threshold rejects more multijet events but also removes more signal. The selected threshold provides substantial multijet rejection with a small signal loss in the studied sample.

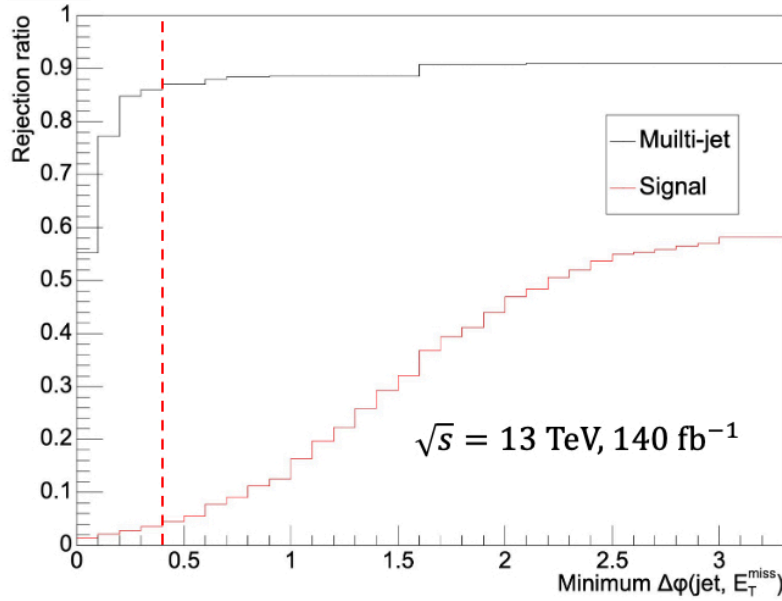


Fig. B.2 Rejected fractions of simulated signal and multijet events as a function of the minimum jet- $\vec{p}_T^{\text{miss}}$ angular threshold X , with $R_{\text{miss}/j} < 6$ also required in the rejected sample. The red and black curves show signal loss and multijet rejection, respectively. The vertical dotted line marks the adopted threshold $X = 0.4$. The denominator is the loose selection in Table B.1.

B.4 Residual multijet contribution

The residual multijet contribution is examined after applying the cleaning cut and the full SR requirements. No simulated multijet events survive those selections in the samples used for the study. This supports neglecting the multijet contribution in the nominal SR prediction, but zero surviving simulated events does not establish an exactly zero physical rate or a quantitative upper bound without the effective sample size.

Jets misidentified as τ candidates can still occur in other background processes. In particular, $Z \rightarrow \nu\nu$ production with jets contains genuine missing momentum and can pass the multijet cleaning even when the selected τ candidate is a jet. This contribution is included in the simulated Z+jets background. A supplementary selection requiring Medium τ identification while failing Tight also gives a negligible multijet contribution in the simulated study, supporting the effectiveness of the cleaning beyond the nominal Tight selection.

Appendix C

Zero- b -jet signal-region optimisation

The $E_{\text{T}}^{\text{miss}}$ threshold in SR0b is studied to assess whether tighter selections can improve the sensitivity at large β_L^{23} . A higher threshold can suppress both the SM background and the destructive interference, but also reduces the retained signal. The significance is therefore evaluated as the threshold is varied, including only the leading systematic uncertainties.

The threshold is scanned from 200 to 1500 GeV in 100 GeV steps. The signal parameter points are $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 0.5, 1.8)$ for SR0b-Res and $(3.0 \text{ TeV}, 2.0, 1.8)$ for SR0b-NonRes. The large value of β_L^{23} makes these benchmarks relevant to the zero- b -jet categories discussed in Section 7.3. The scan outcomes and adopted thresholds are summarised in Table C.1.

Table C.1 Signal benchmarks and preferred thresholds in the supplementary SR0b $E_{\text{T}}^{\text{miss}}$ scan. The last column gives the thresholds used in the final analysis.

Region	$(M_U [\text{TeV}], g_U, \beta_L^{23})$	Scan maximum	Adopted $E_{\text{T}}^{\text{miss}}$ cut
SR0b-Res	(1.5, 0.5, 1.8)	600 GeV	> 200 GeV
SR0b-NonRes	(3.0, 2.0, 1.8)	400 GeV	> 400 GeV

For SR0b-NonRes, the significance is maximised at $E_{\text{T}}^{\text{miss}} > 400 \text{ GeV}$. The scan therefore supports retaining the threshold used in the final region definition. For SR0b-Res, the maximum occurs at $E_{\text{T}}^{\text{miss}} > 600 \text{ GeV}$, giving an improvement of approximately 20% in the significance at the chosen benchmark. The final resonant selection nevertheless retains $E_{\text{T}}^{\text{miss}} > 200 \text{ GeV}$ to keep its kinematic requirements aligned with SR1b-Res. The higher threshold also reduces the magnitude of the destructive interference in SR0b-Res. This demonstrates the role of $E_{\text{T}}^{\text{miss}}$ in suppressing both the SM background and the interference contribution, although the gain must be balanced against the loss of signal events.

References

- [1] Motoi Endo, Syuhei Iguro, Teppei Kitahara, Michihisa Takeuchi, and Ryoutaro Watanabe. Non-resonant new physics search at the lhc for the $b \rightarrow c\tau\nu$ anomalies. *JHEP*, 02:106, 2022. doi: 10.1007/JHEP02(2022)106.
- [2] ATLAS Collaboration. Formulae for estimating significance. ATL-PHYS-PUB-2020-025, 2020. URL <https://cds.cern.ch/record/2736148>.